



PHP Practicum



LARAVEL FRAMEWORK

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Laravel Framework

- Laravel is an open-source PHP framework, which is robust and easy to understand.
- It follows a model-view-controller design pattern.
- Laravel reuses the existing components of different frameworks which helps in creating a web application.
- The web application thus designed is more structured and pragmatic.
- Laravel offers a rich set of functionalities which incorporates the basic features of PHP frameworks like CodeIgniter, Yii and other programming languages like Ruby on Rails.
- Laravel has a very rich set of features which will boost the speed of web development.



Advantages of Laravel

- Elegant Syntax and Developer-Friendly.
- Comprehensive Documentation and Strong Community Support.
- Laravel simplifies the implementation of authentication and authorization.
- Laravel includes a powerful command-line tool called Artisan, which automates repetitive tasks and speeds up development.
- Laravel's Eloquent ORM provides an intuitive and straightforward way to interact with the database.
- Modular Packaging System.



Composer

- Composer is a tool which includes all the dependencies and libraries. It allows a user to create a project with respect to the mentioned framework (for example, those used in Laravel installation). Third party libraries can be installed easily with help of composer.
- Artisan - Command line interface used in Laravel is called Artisan. It includes a set of commands which assists in building a web application.

```
$ composer

Composer version 1.4.1 2017-03-10 09:29:45

Usage:
  command [options] [arguments]

Options:
  -h, --help                Display this help message
  -q, --quiet               Do not output any message
  -V, --version             Display this application version
      --ansi               Force ANSI output
      --no-ansi            Disable ANSI output
  -n, --no-interaction      Do not ask any interactive question
      --profile            Display timing and memory usage information
      --no-plugins         Whether to disable plugins.
  -d, --working-dir=WORKING-DIR If specified, use the given directory as working directory.
  -v|vv|vvv, --verbose      Increase the verbosity of messages: 1 for normal output, 2 for more verbose output and 3 for debug

Available commands:
  about               Short information about Composer.
  archive             Create an archive of this composer package.
  browse             Opens the package's repository URL or homepage in your browser.
  clear-cache         Clears composer's internal package cache.
  clearcache          Clears composer's internal package cache.
  config             Set config options.
  create-project      Create new project from a package into given directory.
  depends            Shows which packages cause the given package to be installed.
  diagnose            Diagnoses the system to identify common errors.
  dump-autoload       Dump the autoloader
```

Laravel Installation - Step 01

- Visit the following URL and download composer to install it on your computer.

<https://getcomposer.org/download/>



Laravel Installation - Step 02

- After the Composer is installed, check the installation by typing the Composer command in the command prompt as shown in the following screenshot.

```
$ composer

Composer version 1.4.1 2017-03-10 09:29:45

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  command [options] [arguments]

Options:
  -h, --help                Display this help message
  -q, --quiet               Do not output any message
  -V, --version             Display this application version
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  diagnose       Diagnoses the system to identify common errors.
  dump-autoload  Dump the autoloader
```

Laravel Installation - Step 03

- Create a new directory anywhere in your system for your new Laravel project. After that, move to path where you have created the new directory and type the following command there to install Laravel.
- Now, we will focus on installation of version 5.7. In Laravel version 5.7, you can install the complete framework by typing the following command.

```
composer create-project laravel/Laravel example-app
```

- The Laravel framework can be directly installed with develop branch which includes the latest framework.



Laravel Installation - Step 04

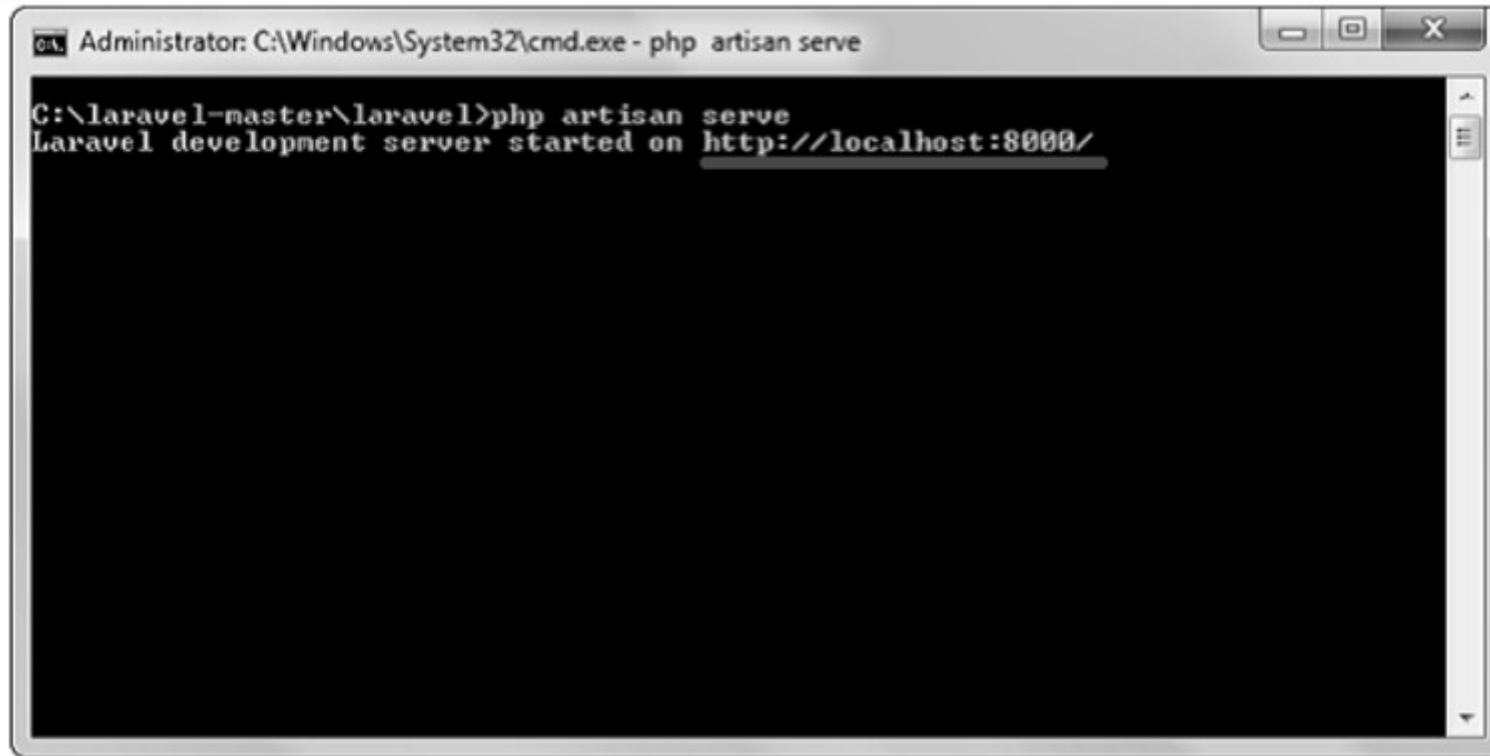
- The above command will install Laravel in the current directory. Start the Laravel service by executing the following command.

```
php artisan serve
```



Laravel Installation - Step 05

- After executing the above command, you will see a screen as shown below,



```
Administrator: C:\Windows\System32\cmd.exe - php artisan serve  
  
C:\laravel-master\laravel>php artisan serve  
Laravel development server started on http://localhost:8000/
```

The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\System32\cmd.exe - php artisan serve". The command prompt displays the command `php artisan serve` being executed, followed by the output message "Laravel development server started on http://localhost:8000/". The URL is underlined in the original image. The window has a standard Windows title bar with minimize, maximize, and close buttons.

Laravel Installation - Step 06

- Copy the URL underlined in gray in the above screenshot and open that URL in the browser. If you see the following screen, it implies Laravel has been installed successfully.



Laravel 5



Activity 01

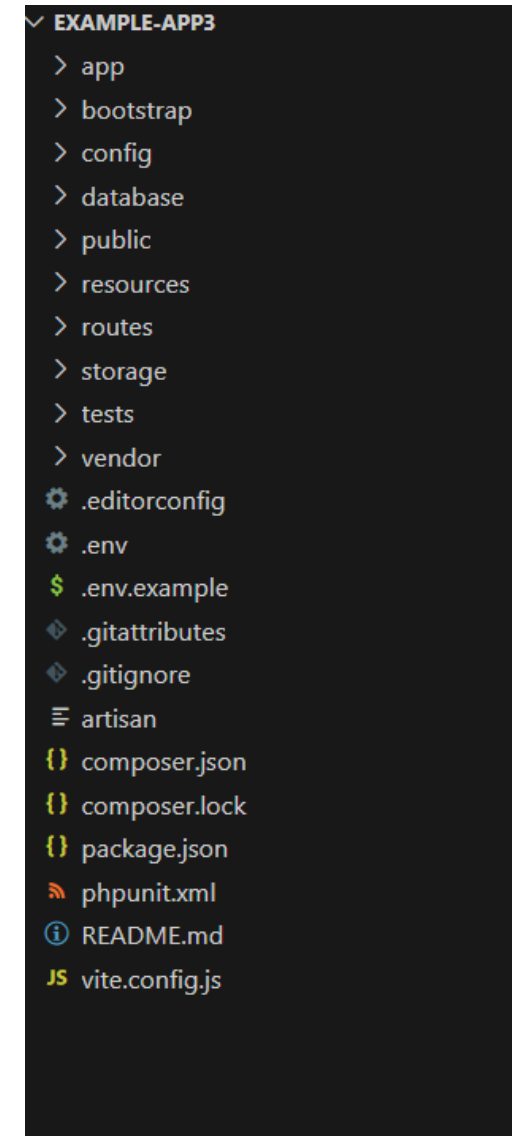
- Create Laravel Project in your computer?



Laravel - Application Structure

The application structure in Laravel is basically the structure of folders, sub-folders and files included in a project. Once we create a project in Laravel, we get an overview of the application structure as shown in the image here.

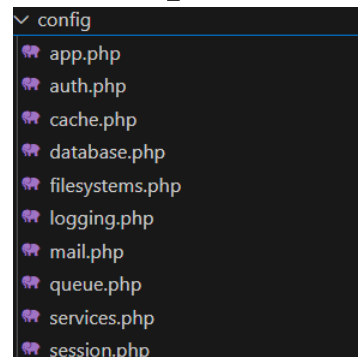
The snapshot shown here refers to the root folder of Laravel namely **Example-APP3**. It includes various sub-folders and files.



```
EXAMPLE-APP3
├── app
├── bootstrap
├── config
├── database
├── public
├── resources
├── routes
├── storage
├── tests
├── vendor
├── .editorconfig
├── .env
├── .env.example
├── .gitattributes
├── .gitignore
├── artisan
├── composer.json
├── composer.lock
├── package.json
├── phpunit.xml
├── README.md
└── vite.config.js
```

Application Folder Structure

- **App** - It is the application folder and includes the entire source code of the project. It contains events, exceptions and middleware declaration.
- **Bootstrap** - This folder encloses all the application bootstrap scripts. It contains a sub-folder namely cache, which includes all the files associated for caching a web application. You can also find the file app.php, which initializes the scripts necessary for bootstrap.
- **Config** - The config folder includes various configurations and associated parameters required for the smooth functioning of a Laravel application. Various files included within the config folder are as shown in the image here. The filenames work as per the functionality associated with them.



Application Folder Structure(Con..)

- **Database** - As the name suggests, this directory includes various parameters for database functionalities. It includes three sub-directories as given below
 - **Seeds** - This contains the classes used for unit testing database.
 - **Migrations** - This folder helps in queries for migrating the database used in the web application.
 - **Factories** - This folder is used to generate large number of data records.

 - **Public** - It is the root folder which helps in initializing the Laravel application. It includes the following files and folders.
 - **.htaccess** - This file gives the server configuration.
 - **javascript and css** - These files are considered as assets.
 - **index.php** - This file is required for the initialization of a web application.
-

Application Folder Structure(Con..)

- **Resources** - Resources directory contains the files which enhances your web application. The sub folders included in this directory and their purpose is explained below,
 - **assets** – The assets folder include files such as LESS and SCSS, that are required for styling the web application.
 - **lang** – This folder includes configuration for localization or internalization.
 - **views** – Views are the HTML files or templates which interact with end users and play a primary role in MVC architecture.
 - **Storage** - This is the folder that stores all the logs and necessary files which are needed frequently when a Laravel project is running. The sub-folders included in this directory and their purpose is given below,
 - **app** – This folder contains the files that are called in succession.
 - **framework** – It contains sessions, cache and views which are called frequently.
 - **Logs** – All exceptions and error logs are tracked in this sub folder.
-

Application Folder Structure(Con..)

- **Tests** - All the unit test cases are included in this directory.
 - **Vendor** - Laravel is completely based on Composer dependencies, for example to install Laravel setup or to include third party libraries, etc. The Vendor folder includes all the composer dependencies. In addition to the above-mentioned files, Laravel also includes some other files which play a primary role in various functionalities such as GitHub configuration, packages and third-party libraries.
-

Laravel - Routing

- In Laravel, all requests are mapped with the help of routes. Basic routing routes the request to the associated controllers.
 - Basic Routing
 - Route parameters
 - Named Routes
-

Basic Routing

- All the application routes are registered within the app/routes.php file. This file tells Laravel for the URIs it should respond to, and the associated controller will give it a particular call.

```
Route::get('/', function () { return view('welcome'); });
```

```
Route::get('/aboutus', function () { return view('aboutus'); });
```

```
Route::get('/contactus', function () { return view('contactus'); });
```

Activity 02

- Create a sample web page file in your project?
- Make routes for the created web page?
- Run the created Laravel web page in your web browser?



Thank you!

